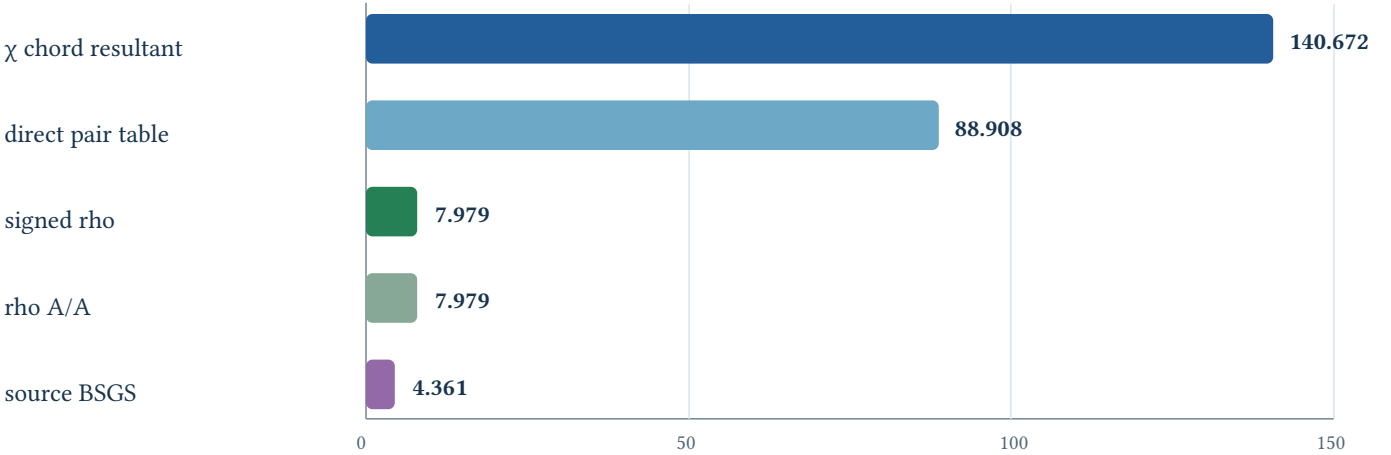
Held-out order-229 complete solves: counted group work

Four targets × three rounds per arm; mean cold S_group = operations / √229



χ and direct-pair rows: identical trials, relations, rank-nine recovery and scalar outputs in 12/12 pairs.

Source: HELDOUT_FIVE_ARM_V2.summary.json; 60 raw children and independent checker receipt.